



# COMPILER VS INTERPRETER

CSA1402-COMPLIER DESIGN

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# INTRODUCTION

- A **compiler** and **interpreter** are language processors.
- They convert high-level programming languages into machine code.
- Both help the computer understand and execute programs.
- The main difference is **how they translate the code**.



# WHAT IS A COMPILER?

- **Definition**

- A **compiler** translates the **entire program at once** into machine code before execution.

- **Features**

- Converts the complete program.
- Produces an executable (.Exe) file.
- Executes faster.
- Errors are shown after compilation.

- **Examples**

- C, C++, go, rust

## Workflow

Source Program



Lexical Analysis



Syntax Analysis



Semantic Analysis



Intermediate Code Generation



Code Optimization



Machine Code Generation



Executable File



Program Execution



# WHAT IS AN INTERPRETER?

- **Definition**

- An **interpreter** translates and executes the program **line by line**.

- **Features**

- No executable file is created.
- Stops immediately if an error occurs.
- Easier to debug.
- Slower execution.

- **Examples**

- Python, javascript, ruby, php

## Workflow

Source Program



Read One Statement



Lexical Analysis



Syntax Analysis



Execute Statement

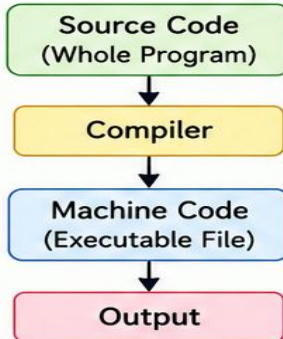


Next Statement



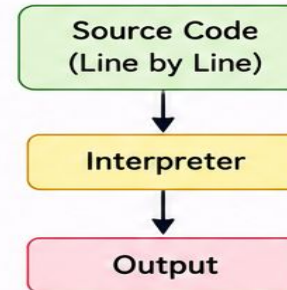
# Compiler vs Interpreter

## COMPILER



- Translates the entire program at once.
- Produces an executable file.
- Errors are shown after compilation.
- Runs faster.
- **Example:** C, C++

## INTERPRETER



- Translates and executes the program line by line.
- No executable file is created.
- Errors are shown immediately.
- Runs slower.
- **Example:** Python, JavaScript

## COMPARISON TABLE

| Feature         | Compiler                            | Interpreter                          |
|-----------------|-------------------------------------|--------------------------------------|
| Translation     | Translates entire program at once.  | Translates line by line.             |
| Executable File | Yes, creates executable file.       | No, does not create executable file. |
| Error Reporting | Errors are shown after compilation. | Errors are shown immediately.        |
| Execution Speed | Faster execution.                   | Slower execution.                    |
| Memory Usage    | More memory required.               | Less memory required.                |
| Debugging       | Harder to debug.                    | Easier to debug.                     |
| Examples        | C, C++                              | Python, JavaScript                   |



**Summary:** Compiler translates the whole program and runs faster. Interpreter translates one line at a time and is easier to debug.





# REAL-LIFE EXAMPLES



## Examination Papers

### Compiler

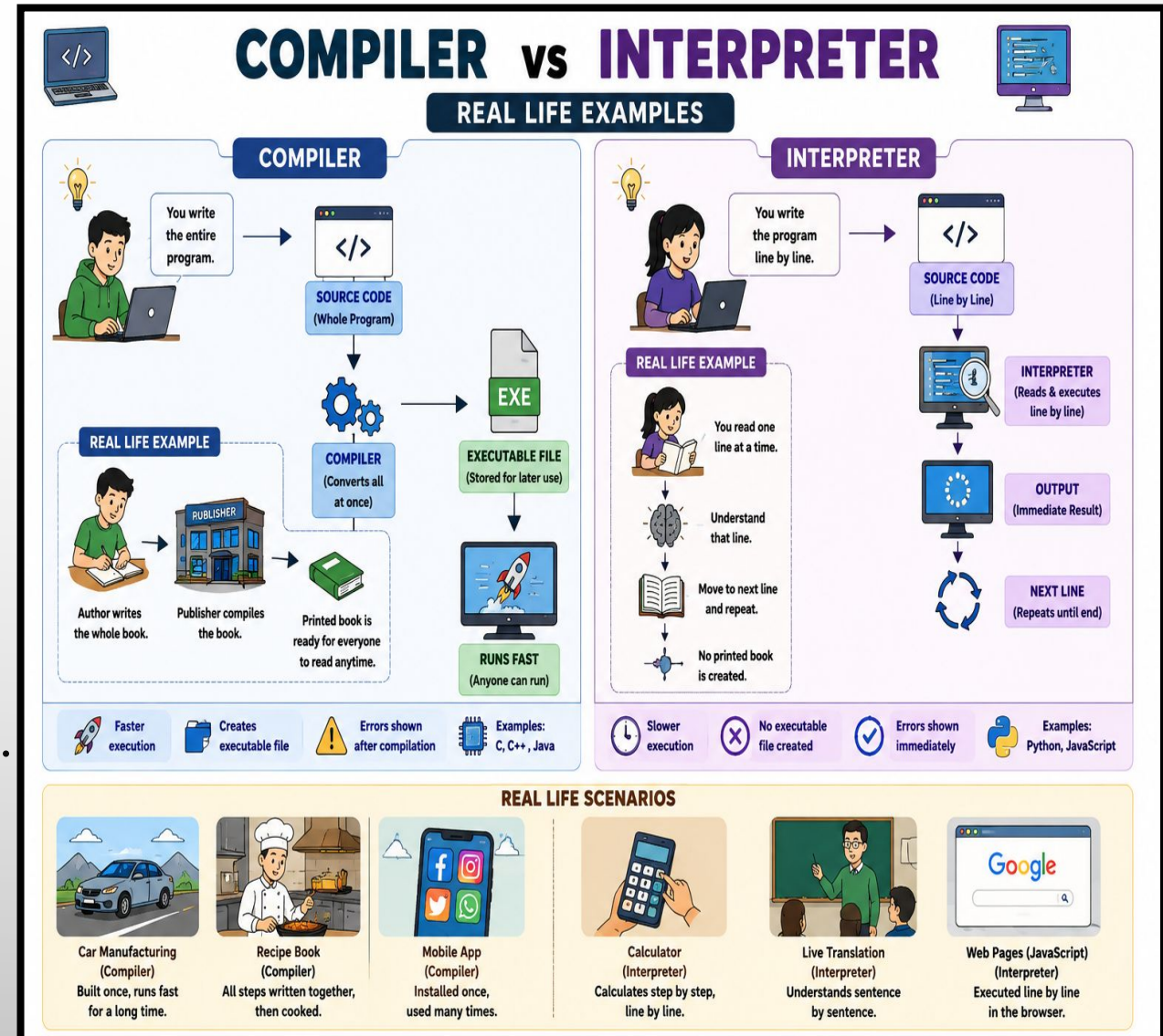
Teacher checks the **entire answer sheet**.

- Finds every mistake.
- Gives all corrections together.

### Interpreter

Teacher checks **one answer at a time**.

- If Question 2 is missing, checking stops.
- Student fixes it before continuing.





# ADVANTAGES & DISADVANTAGES

- **Compiler**
- **Advantages**
  - Faster execution
  - Better performance
  - Suitable for large projects
- **Disadvantages**
  - Compilation takes time
  - Harder to debug

- **Interpreter**
- **Advantages**
  - Easy to debug
  - Executes immediately
  - Good for beginners
- **Disadvantages**
  - Slower execution
  - No executable file



# COMPLIER

| main.c                                                                                                         | Run | Output                                                         |
|----------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------|
| <pre>1 #include &lt;stdio.h&gt; 2 3 int main() { 4     printf("Hello, Compiler!\n"); 5     return 0; 6 }</pre> |     | <pre>Hello, Compiler!  === Code Execution Successful ===</pre> |

# INTERPRETER

| main.py                                   | Run | Output                                                            |
|-------------------------------------------|-----|-------------------------------------------------------------------|
| <pre>1 print("Hello, Interpreter!")</pre> |     | <pre>Hello, Interpreter!  === Code Execution Successful ===</pre> |





# CONCLUSION

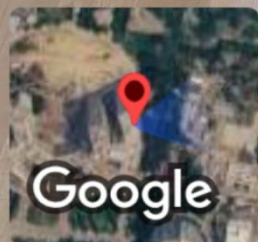
- Both compiler and interpreter convert high-level language into machine language.
- A **compiler** translates the whole program before execution.
- An **interpreter** translates one line at a time.
- The choice depends on the programming language and application requirements.



# THANK YOU



GPS Map Camera



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